

Year 7 • Chapter 7 • Decimals: answers (teacher copy)

7.01 Ordering decimals (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: 7.48 or 7.9 $\rightarrow$ 7.9
- Column 1, E2: 9.59 or 9.56 $\rightarrow$ 9.59
- Column 2, E1: 4.5, 4.38, 4.35, 4.24 $\rightarrow$ 4.500, 4.380, 4.350, 4.240; 4.24, 4.35, 4.38, 4.5
- Column 2, E2: 4.5, 4.47, 4.673, 4.27 $\rightarrow$ 4.500, 4.470, 4.673, 4.270; 4.27, 4.47, 4.5, 4.673
- Column 3, E1: Write a decimal between 2.5 and 2.6. $\rightarrow$ 2.50 and 2.60; e.g. 2.55 is between them
- Column 3, E2: Write a decimal between 1.8 and 1.9. $\rightarrow$ 1.80 and 1.90; e.g. 1.85 is between them

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------|-----------------------|-----------------------|
| 1. 9.14 | 20. 6.71 | 34. 3.277, 3.49, 3.5, |
| 2. 3.13 | 21. 7.5 | 3.56 |
| 3. 9.16 | 22. 8.88 | 35. 4.14, 4.3, 4.39, |
| 4. 3.78 | 23. 9.43 | 4.91 |
| 5. 1.8 | 24. 2.25 | 36. 2.24, 2.293, 2.5, |
| 6. 8.85 | 25. 4.93 | 2.58 |
| 7. 4.32 | 26. 5.46 | 37. 5.06, 5.236, 5.4, |
| 8. 1.22 | 27. 5.76 | 5.75 |
| 9. 9.92 | 28. 10.56 | 38. 5.54, 5.553, 5.6, |
| 10. 5.7 | 29. 8.4 | 5.9 |
| 11. 3.31 | 30. 5.36, 5.7, 5.825, | 39. 0.6, 0.73, 0.74, |
| 12. 3.3 | 5.88 | 0.948 |
| 13. 5.56 | 31. 3.2, 3.22, 3.635, | 40. e.g. 5.65 |
| 14. 5.9 | 3.9 | 41. e.g. 3.15 |
| 15. 1.27 | 32. 1.1, 1.271, 1.52, | 42. e.g. 4.65 |
| 16. 9 | 1.56 | 43. e.g. 7.65 |
| 17. 9.81 | 33. 3.3, 3.36, 3.78, | 44. e.g. 7.25 |
| 18. 5.56 | 3.927 | 45. e.g. 8.25 |
| 19. 7.23 | | 46. e.g. 6.35 |
| | | 47. e.g. 5.55 |

7.02 Decimals and fractions (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: 0.317 $\rightarrow \frac{317}{1000}$
- Column 1, E2: 0.754 $\rightarrow \frac{754}{1000}$
- Column 2, E1: 0.08 $\rightarrow = \frac{8}{100} = \frac{2}{25}$
- Column 2, E2: 0.72 $\rightarrow = \frac{72}{100} = \frac{18}{25}$
- Column 3, E1: 8.16 $\rightarrow = 8$ and $\frac{16}{100} = 8 \frac{4}{25}$
- Column 3, E2: 4.02 $\rightarrow = 4$ and $\frac{2}{100} = 4 \frac{1}{50}$

Questions (in order: Easy, Medium, Challenging)

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|-----------------------|------------------------|------------------------|
| 1. $\frac{628}{1000}$ | 10. $\frac{874}{1000}$ | 19. $\frac{3}{10}$ |
| 2. $\frac{52}{100}$ | 11. $\frac{10}{100}$ | 20. $\frac{977}{1000}$ |
| 3. $\frac{2}{10}$ | 12. $\frac{4}{10}$ | 21. $\frac{67}{100}$ |
| 4. $\frac{49}{100}$ | 13. $\frac{781}{1000}$ | 22. $\frac{6}{10}$ |
| 5. $\frac{39}{100}$ | 14. $\frac{873}{1000}$ | 23. $\frac{46}{100}$ |
| 6. $\frac{8}{10}$ | 15. $\frac{558}{1000}$ | 24. $\frac{48}{100}$ |
| 7. $\frac{456}{1000}$ | 16. $\frac{213}{1000}$ | 25. $\frac{858}{1000}$ |
| 8. $\frac{7}{10}$ | 17. $\frac{8}{100}$ | 26. $\frac{154}{1000}$ |
| 9. $\frac{98}{100}$ | 18. $\frac{28}{100}$ | 27. $\frac{45}{100}$ |

- | | | |
|------------------------|---------------------|-----------------------|
| 28. $\frac{37}{100}$ | 37. $\frac{3}{50}$ | 46. $4 \frac{27}{50}$ |
| 29. $\frac{623}{1000}$ | 38. $\frac{9}{20}$ | 47. $2 \frac{47}{50}$ |
| 30. $\frac{3}{4}$ | 39. $\frac{11}{20}$ | 48. $5 \frac{16}{25}$ |
| 31. $\frac{23}{50}$ | 40. $\frac{11}{50}$ | 49. $2 \frac{9}{20}$ |
| 32. $\frac{41}{50}$ | 41. $\frac{4}{25}$ | 50. $6 \frac{9}{25}$ |
| 33. $\frac{22}{25}$ | 42. $\frac{33}{50}$ | 51. $2 \frac{22}{25}$ |
| 34. $\frac{1}{20}$ | 43. $\frac{7}{50}$ | 52. $9 \frac{4}{25}$ |
| 35. $\frac{17}{20}$ | 44. $\frac{14}{25}$ | |
| 36. $\frac{39}{50}$ | 45. $\frac{23}{25}$ | |

7.03 Adding and subtracting decimals

(booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: 7.3 – 5.8 $\rightarrow$ 1.5
- Column 1, E2: 7.7 – 6.5 $\rightarrow$ 1.2
- Column 2, E1: 4.9 – 3.79 $\rightarrow$ 4.90 – 3.79; = 1.11
- Column 2, E2: 69 – 3.94 $\rightarrow$ 69.00 – 3.94; = 65.06
- Column 3, E1: Mia runs 3.26 km and then 4.39 km. How far does she run altogether? $\rightarrow$ 3.26 + 4.39; = 7.65 km
- Column 3, E2: A 12 m rope has 3.75 m cut off. How much is left? $\rightarrow$ 12.00 – 3.75; = 8.25 m

Questions (in order: Easy, Medium, Challenging)

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|----------|-----------|--------------|
| 1. 7.8 | 21. 0.4 | 41. 38.65 |
| 2. 3.7 | 22. 11.4 | 42. 2.8 |
| 3. 0.3 | 23. 11.6 | 43. 37.62 |
| 4. 13.7 | 24. 10.5 | 44. 0.58 |
| 5. 1.9 | 25. 11.1 | 45. 4.97 |
| 6. 13.7 | 26. 14.8 | 46. 11.59 |
| 7. 8.6 | 27. 17.5 | 47. 20.29 |
| 8. 13.6 | 28. 3.8 | 48. 3.88 |
| 9. 10.8 | 29. 11 | 49. 24.44 |
| 10. 9.6 | 30. 13.8 | 50. 10.75 |
| 11. 14.6 | 31. 6.22 | 51. 7.04 |
| 12. 13.5 | 32. 5.55 | 52. 71.83 |
| 13. 4.3 | 33. 19.61 | 53. 14.96 km |
| 14. 3.1 | 34. 32.03 | 54. 10.23 m |
| 15. 16.1 | 35. 34.44 | 55. 6.16 km |
| 16. 12.4 | 36. 29.11 | 56. 8.18 m |
| 17. 13.4 | 37. 78.78 | 57. 5.31 km |
| 18. 0.7 | 38. 9.57 | 58. 6.65 m |
| 19. 14 | 39. 8.19 | 59. 12.14 km |
| 20. 11.9 | 40. 57.5 | |

7.04 Multiplying and dividing by powers of 10 (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: 5.15 $\times$ 100 $\rightarrow$ 515
- Column 1, E2: 0.769 $\times$ 10 $\rightarrow$ 7.69
- Column 1, E3: 583.3 $\div$ 1 000 $\rightarrow$ 0.5833
- Column 2, E1: 7.84 $\div$ $\square$ = 0.00784 $\rightarrow$ moved 3 places right; $\square$ = 1 000
- Column 2, E2: 4.68 $\times$ $\square$ = 4 680 $\rightarrow$ moved 3 places left; $\square$ = 1 000
- Column 3, E1: 99.9 m to centimetres $\rightarrow$ 1 m = 100 cm, so $\times$ 100; 99.9 $\times$ 100 = 9 990 cm

7. Column 3, E2: 203 g to kilograms $\rightarrow 1 \text{ kg} = 1000 \text{ g}$, so $\div 1000$; $203 \div 1000 = 0.203 \text{ kg}$
 8. Column 3, E3: 71.3 cm to millimetres $\rightarrow 1 \text{ cm} = 10 \text{ mm}$, so $\times 10$; $71.3 \times 10 = 713 \text{ mm}$

Questions (in order: Easy, Medium, Challenging)

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|------------|-----------|--------------|
| 1. 5 660 | 15. 35.22 | 29. 1 000 |
| 2. 6.66 | 16. 39.43 | 30. 1 000 |
| 3. 34 | 17. 959.3 | 31. 100 |
| 4. 8 550 | 18. 41.68 | 32. 100 |
| 5. 22.6 | 19. 100.5 | 33. 100 |
| 6. 6.74 | 20. 55.97 | 34. 8 830 cm |
| 7. 150 | 21. 94.89 | 35. 0.343 kg |
| 8. 86.9 | 22. 49.58 | 36. 502 mm |
| 9. 6 710 | 23. 3.375 | 37. 3 530 cm |
| 10. 73.2 | 24. 5.844 | 38. 0.495 kg |
| 11. 23.9 | 25. 10 | 39. 95 mm |
| 12. 427 | 26. 1 000 | 40. 2 980 cm |
| 13. 0.8821 | 27. 1 000 | 41. 0.157 kg |
| 14. 0.1727 | 28. 10 | |

7.05 Using a known product (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: $6 \times 0.07 \rightarrow 0.42$
- Column 1, E2: $0.6 \times 0.7 \rightarrow 0.42$
- Column 1, E3: $0.6 \times 0.06 \rightarrow 0.036$
- Column 2, E1: $21 \times 48 = 1\ 008$. Find 2.1×4.8 . $\rightarrow 2$ decimal places; $= 10.08$
- Column 2, E2: $33 \times 34 = 1\ 122$. Find 0.33×3.4 . $\rightarrow 3$ decimal places; $= 1.122$
- Column 3, E1: $25 \times 45 = 1\ 125$. Find $112.5 \div 45$. $\rightarrow 1\ 125 \div 45 = 25$; 112.5 is $1125 \div 10$, so the answer is $25 \div 10$; $= 2.5$
- Column 3, E2: $13 \times 17 = 221$. Find $2.21 \div 17$. $\rightarrow 221 \div 17 = 13$; 2.21 is $221 \div 100$, so the answer is $13 \div 100$; $= 0.13$

Questions (in order: Easy, Medium, Challenging)

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|-----------|-----------|-----------|
| 1. 0.0042 | 14. 0.12 | 27. 0.812 |
| 2. 4.2 | 15. 0.063 | 28. 4.65 |
| 3. 0.042 | 16. 0.22 | 29. 79.9 |
| 4. 0.042 | 17. 0.27 | 30. 70.2 |
| 5. 2.4 | 18. 5 | 31. 155.1 |
| 6. 0.55 | 19. 1.4 | 32. 86.4 |
| 7. 0.054 | 20. 0.33 | 33. 4.2 |
| 8. 0.08 | 21. 3.3 | 34. 0.36 |
| 9. 0.36 | 22. 0.04 | 35. 1.3 |
| 10. 0.18 | 23. 0.08 | 36. 3.4 |
| 11. 0.2 | 24. 0.05 | 37. 4.8 |
| 12. 0.02 | 25. 0.532 | 38. 4.2 |
| 13. 0.016 | 26. 25.6 | 39. 0.43 |

7.06 Multiplying decimals (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: $9.3 \times 5 \rightarrow 46.5$
- Column 1, E2: $3.5 \times 2 \rightarrow 7$
- Column 2, E1: $0.42 \times 0.6 \rightarrow 42 \times 6 = 252$; 3 decimal places: 0.252
- Column 2, E2: $0.3 \times 0.5 \rightarrow 30 \times 5 = 150$; 3 decimal places: 0.15
- Column 3, E1: $8.47 \times 9.5 \rightarrow$ estimate: $8 \times 10 = 80$; $847 \times 95 = 80\ 465$; 3 decimal places: 80.465
- Column 3, E2: $3.74 \times 2.1 \rightarrow$ estimate: $4 \times 2 = 8$; $374 \times 21 = 7\ 854$; 3 decimal places: 7.854

Questions (in order: Easy, Medium, Challenging)

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|---------|---------|---------|
| 1. 26.6 | 4. 21.3 | 7. 13.5 |
| 2. 26.5 | 5. 28.8 | 8. 48 |
| 3. 24 | 6. 16.1 | 9. 22.8 |

- | | | |
|----------|-----------|------------|
| 10. 67.2 | 23. 10.2 | 36. 0.198 |
| 11. 64.4 | 24. 16 | 37. 0.028 |
| 12. 36 | 25. 16.4 | 38. 0.234 |
| 13. 63 | 26. 15.8 | 39. 0.2 |
| 14. 71.1 | 27. 60.8 | 40. 0.416 |
| 15. 31.6 | 28. 11.8 | 41. 58.05 |
| 16. 27.2 | 29. 62.3 | 42. 62.752 |
| 17. 21.6 | 30. 12.8 | 43. 23.636 |
| 18. 49.5 | 31. 0.036 | 44. 53.246 |
| 19. 84.6 | 32. 0.513 | 45. 22.673 |
| 20. 48.3 | 33. 0.774 | 46. 8.211 |
| 21. 20.4 | 34. 0.54 | 47. 9.984 |
| 22. 58.1 | 35. 0.342 | |

7.07 Dividing decimals by whole numbers

(booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: $13.3 \div 7 \rightarrow 1.9$
- Column 1, E2: $9.6 \div 6 \rightarrow 1.6$
- Column 2, E1: $4.6 \div 2 \rightarrow 4.600 \div 2$; $= 2.3$
- Column 2, E2: $7 \div 5 \rightarrow 7.000 \div 5$; $= 1.4$
- Column 3, E1: \$16.30 is shared equally between 5 people. How much does each get? $\rightarrow \$16.30 \div 5$; $= \$3.26$ each
- Column 3, E2: \$58.92 is shared equally between 4 people. How much does each get? $\rightarrow \$58.92 \div 4$; $= \$14.73$ each

Questions (in order: Easy, Medium, Challenging)

- | | | |
|---------|------------|-------------|
| 1. 3.1 | 21. 7.3 | 41. 2.05 |
| 2. 5.4 | 22. 1.8 | 42. 0.2875 |
| 3. 3 | 23. 9.6 | 43. 0.275 |
| 4. 9.4 | 24. 1.3 | 44. 0.175 |
| 5. 4.7 | 25. 3.5 | 45. 0.26 |
| 6. 8 | 26. 1.5 | 46. 0.7 |
| 7. 3.3 | 27. 3.6 | 47. 1.35 |
| 8. 9.3 | 28. 5.9 | 48. 1.16 |
| 9. 4.8 | 29. 5.5 | 49. 1.675 |
| 10. 9.8 | 30. 8.3 | 50. 0.9 |
| 11. 9.9 | 31. 1.9 | 51. 1.575 |
| 12. 1.9 | 32. 0.9 | 52. 1.8 |
| 13. 4 | 33. 0.85 | 53. \$17.20 |
| 14. 4.8 | 34. 1.05 | 54. \$8.90 |
| 15. 2.5 | 35. 1.35 | 55. \$15.98 |
| 16. 1.2 | 36. 2.35 | 56. \$17.43 |
| 17. 4.1 | 37. 0.95 | 57. \$1.80 |
| 18. 8.6 | 38. 0.6125 | 58. \$16.26 |
| 19. 2.6 | 39. 0.4875 | |
| 20. 1.5 | 40. 0.65 | |

7.08 Dividing by a decimal (booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: $22.5 \div 0.9 \rightarrow 25$
- Column 1, E2: $23.4 \div 0.9 \rightarrow 26$
- Column 2, E1: $4.05 \div 0.5 \rightarrow 40.5 \div 5$; $= 8.1$
- Column 2, E2: $4.3 \div 0.5 \rightarrow 43 \div 5$; $= 8.6$
- Column 3, E1: A bottle holds 0.25 L. How many bottles can be filled from 4.5 L? $\rightarrow 4.5 \div 0.25$; $= 450 \div 25$; $= 18$ bottles
- Column 3, E2: A bottle holds 0.25 L. How many bottles can be filled from 1 L? $\rightarrow 1 \div 0.25$; $= 100 \div 25$; $= 4$ bottles

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------|--------|--------|
| 1. 28 | 7. 26 | 13. 17 |
| 2. 8 | 8. 27 | 14. 27 |
| 3. 9 | 9. 11 | 15. 19 |
| 4. 26 | 10. 27 | 16. 10 |
| 5. 16 | 11. 3 | 17. 21 |
| 6. 5 | 12. 23 | 18. 30 |

19. 13	32. 6.6	45. 2.7
20. 4	33. 8.7	46. 3.8
21. 8	34. 6.8	47. 2.1
22. 22	35. 9	48. 8.4
23. 28	36. 6	49. 8.3
24. 26	37. 1.5	50. 2.2
25. 22	38. 4.1	51. 1.4
26. 26	39. 8.3	52. 8 bottles
27. 25	40. 9.5	53. 16 bottles
28. 24	41. 2.6	54. 6 bottles
29. 2	42. 1.2	55. 15 bottles
30. 4.4	43. 6.1	56. 24 bottles
31. 2.5	44. 5	

7.09 Fractions to decimals (booklet p. 11)

WE DO examples (teacher)

- Column 1, E1: $\frac{18}{25} \rightarrow 0.72$
- Column 1, E2: $\frac{1}{10} \rightarrow 0.1$
- Column 2, E1: $\frac{7}{16} \rightarrow 7.0000 \div 16; = 0.4375$
- Column 2, E2: $\frac{1}{8} \rightarrow 1.0000 \div 8; = 0.125$
- Column 3, E1: $\frac{1}{3} \rightarrow 1 \div 3 = 0.333...; = 0.3$
- Column 3, E2: $\frac{2}{3} \rightarrow 2 \div 3 = 0.666...; = 0.6$

Questions (in order: Easy, Medium, Challenging)

1. 0.42	20. 0.04	39. 0.375
2. 0.4	21. 0.78	40. 0.5625
3. 0.02	22. 0.85	41. 0.9375
4. 0.75	23. 0.8	42. 0.825
5. 0.7	24. 0.3	43. 0.875
6. 0.5	25. 0.45	44. 0.725
7. 0.6	26. 0.36	45. 0.975
8. 0.25	27. 0.82	46. 0.575
9. 0.2	28. 0.46	47. 0.025
10. 0.12	29. 0.05	48. 0.275
11. 0.55	30. 0.675	49. 0.325
12. 0.84	31. 0.0625	50. 0.1
13. 0.68	32. 0.3125	51. 0.5
14. 0.9	33. 0.6875	52. 0.16
15. 0.95	34. 0.925	53. 0.83
16. 0.06	35. 0.525	54. 0.7
17. 0.65	36. 0.625	55. 0.09
18. 0.64	37. 0.8125	56. 1.3
19. 0.54	38. 0.425	57. 0.18

7.10 Rounding decimals (booklet p. 12)

WE DO examples (teacher)

- Column 1, E1: 77.26 $\rightarrow$ 77.3
- Column 1, E2: 89.54 $\rightarrow$ 89.5
- Column 2, E1: 22.0996 (1 d.p.) $\rightarrow$ next digit is 9: round up; = 22.1

- Column 2, E2: 91.2065 (2 d.p.) $\rightarrow$ next digit is 6: round up; = 91.21
- Column 3, E1: $31 \div 7 \rightarrow$ calculator: 4.42857...; = 4.43 (2 d.p.)
- Column 3, E2: $13 \div 3 \rightarrow$ calculator: 4.33333...; = 4.33 (2 d.p.)

Questions (in order: Easy, Medium, Challenging)

1. 81.1	18. 37.4	35. 98.4
2. 70.1	19. 47.7	36. 30.09
3. 42.9	20. 58.0	37. 59.170
4. 1.2	21. 27.2	38. 63.2
5. 13.8	22. 46.2	39. 33.5
6. 33.6	23. 99.9	40. 3.23
7. 65.3	24. 15.1	41. 2.71
8. 48.0	25. 97.1	42. 2.13
9. 19.5	26. 34.6	43. 5.43
10. 40.0	27. 71.6	44. 3.85
11. 86.1	28. 16.5	45. 3.63
12. 36.0	29. 90.0	46. 2.69
13. 74.5	30. 12.878	47. 1.85
14. 86.4	31. 97.32	48. 4.88
15. 25.5	32. 84.504	49. 1.31
16. 69.2	33. 12.1	
17. 27.7	34. 26.16	

7.11 Decimals and money (booklet p. 13)

WE DO examples (teacher)

- Column 1, E1: \$1.18 + \$1.31 $\rightarrow$ \$2.49
- Column 1, E2: \$12.01 + \$17.63 $\rightarrow$ \$29.64
- Column 1, E3: \$1.77 from \$10 $\rightarrow$ \$8.23
- Column 2, E1: 6 items at \$10.82 each $\rightarrow 6 \times \$10.82; = \64.92
- Column 2, E2: 8 items at \$7.68 each $\rightarrow 8 \times \$7.68; = \61.44
- Column 3, E1: You buy 3 drinks at \$5.90 and a sandwich at \$5.01. You pay with \$50. Find the change. $\rightarrow 3 \times \$5.90 = \17.70 ; total: \$17.70 + \$5.01 = \$22.71; change: \$50 - \$22.71 = \$27.29
- Column 3, E2: Round \$70.71 to the nearest 5 cents for a cash payment. $\rightarrow$ cents: 1 $\rightarrow$ 0; = \$70.70

Questions (in order: Easy, Medium, Challenging)

1. \$19.84	15. \$16.60	29. \$54.72
2. \$23.74	16. \$39.48	30. \$88.38
3. \$31.06	17. \$6.69	31. \$11.41
4. \$17.61	18. \$1.27	32. \$50.68
5. \$27.53	19. \$12.73	33. \$74.90
6. \$14.78	20. \$6.80	34. \$69.44
7. \$17.24	21. \$7.96	35. \$70.20
8. \$8.05	22. \$5.37	36. \$9.47
9. \$30.71	23. \$9.10	37. \$60.65
10. \$12.59	24. \$16.18	38. \$28.77
11. \$28.06	25. \$6.77	39. \$78.90
12. \$28.88	26. \$25.26	40. \$28.90
13. \$44.33	27. \$27.10	41. \$16.90
14. \$3.91	28. \$24.50	